Dharmik Umeshkumar Jha

Ahmedabad, Gujarat, India | dharmikjha.xd@gmail.com | 23bme016@nirmauni.ac.in | +91 93166 78961 | linkedin

Portfolio

EDUCATION

Nirma University, Institute of Technology, Ahmedabad, Gujarat, India. *2023 – 2027*

- **Major:** B.Tech Mechanical Engineering (7th Sem) — CGPA: 8.64/10.0
- **Minor:** Data Science
- **Relevant Coursework:** Machine Design, Finite Element Analysis, Design & Dynamics of Machines, Aerodynamics, Manufacturing Processes

EXPERIENCE

Mechanical Design Intern — Shree Siddhi Vinayak Engg. Co. *May 2026 – July 2026*

Industrial Internship

Society of Automotive Engineers (SAE) Nirma Collegiate Club — Team Arrow *Oct 2023 – Mar 2026*

Core Member (2023) → Design Lead (2025) → Captain (2025–2026)

- Led mechanical design across 5 UAV programs (3 national, 2 international competitions) spanning structural design, CAD, aerodynamic sizing, fabrication, and validation; incl. **4th overall + 2nd Best Design Report** at DDC 2024.
- Directed a 9-member team at SAEISS DDC 2025; led design-build-test of a fixed-wing UAV (<4 kg empty weight, 6 kg payload capacity, and a 500 g airdrop mechanism) across a 5-month cycle. Designed a 250 g carbon-fibre primary landing gear (∼2.5% of AUW) with spring-based impact absorption, validated via drop testing. **1st place, Regular Class; 3rd, Micro Class.**
- Led structural/mechanical development of an 800 mm hybrid tilt-rotor tricopter for SAE Aero East 2026: propulsion sizing, thrust-vectoring mechanism (110° rotation), lightweight structural design for autonomous hover-to-cruise transition.
- Designed sub-2 kg quadrotor airframe (Fusion 360 generative design, single Nylon-12 monocoque) for Aerothon 2025: 1.847 kg AUW, 3.86 thrust-to-weight, 21.57 min hover endurance.
- Designed/fabricated 1.6 kg micro-class UAV (SAE Aero West 2025): payload fraction >0.5 under 450 W limit, <10 ft takeoff, 5-min endurance. **11th/75 international teams; 5th in mission score.**

RESEARCH PUBLICATION

- **Journal Paper:** *"Design and Aerodynamic Performance Assessment of High Payload Capacity Fixed-Wing Unmanned Aerial Vehicle (UAV)."*
  the Journal of *Flow Visualization and Image Processing, **Begell House**.*
  DOI: 10.1615/JFlowVisImageProc.2026064024
- **Conference Paper:** *"Design Evolution and Aerodynamic Performance Assessment of Regular-Class Fixed-Wing Unmanned Aerial Vehicle (UAV)."*
  Presented at *12th International & 52nd National Conference on Fluid Mechanics and Fluid Power (FMFP 2025)*, Nirma University.

SIGNIFICANT ACADEMIC PROJECTS

Full write-ups & additional projects: portfolio

Two-Stage Helical Gear Reduction Gearbox *March 2026 – April 2026*

- Designed a two-stage helical gearbox, sizing gears using Lewis bending and wear strength criteria.
- Designed shafts under combined loading using fatigue analysis and DE-Goodman criterion; selected bearings based on load capacity and L10 life.
- Developed the complete SolidWorks assembly and manufacturing drawings with fits, tolerances, and DFM considerations.

Design and Development of Advanced Class UAV – SAE Aero Design East 2026 *Sept 2025 – March 2026*

- Designed tilt-rotor tricopter (800 mm wingspan, 1.6 kg AUW, S1223 + NACA 9325 lifting-body fuselage) capable of autonomous hover-to-cruise transition; mechanically actuated thrust-vectoring system with 110° rotation of the front thrust vectors to enable hover-to-cruise transition.

Aerothon 2025 – Disaster Response Quadrotor UAV *March 2025 – August 2025*

- Designed a sub-2 kg quadrotor (440 mm wheelbase) with a single integrated airframe using **Generative Design**, optimising strength-to-weight ratio and manufacturability; fabricated in 3D-printed PETG.

TECHNICAL SKILLS

- **Design & Analysis:** SolidWorks, Fusion 360 (Generative Design), Assembly Design, ANSYS Structural, Finite Element Analysis
- **Manufacturing, Programming & Flight Systems:** 3D Printing, Laser Cutting, Composite Fabrication, Python (NumPy, Matplotlib), MATLAB, Pixhawk Autopilot, Mission Planner